Hyeonjun Kil

Backend Engineer / Platform Part Leader

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PROFESSIONAL SUMMARY

Backend engineer designing and operating shared backend platforms across 6 live game environments (12 branches), currently leading the Platform Part across real-time game chat, marketing data pipelines, and common package systems. At GameDuo, reduced BigQuery read cost per TB by 82% via Storage Read API migration, expanded pipeline coverage from 60 to 360 days through serverless migration, validated capacity for a real-time chat server targeting 6,000 CCU, and presented the case study at Games on AWS 2025. Over 4 years and 10 months across education, healthcare, clinical trials, and gaming, I have consistently delivered DB optimization, concurrency control, legacy modernization, and event-driven architecture. Recently I have integrated AI tools into an engineering quality system that verifies outputs through TDD, integration tests, review gates, and evidence search rather than unchecked code generation.

CORE COMPETENCIES

DB Optimization, Concurrency & Distributed Systems

- Normalized and indexed a 100+ column marketing table, reducing query latency from 18s to 0.5s
- Deadlock root cause analysis and distributed locking — diagnosing and resolving data-layer bottlenecks
- Advanced Redis Streams WAL with a 4-shard Valkey cluster and serverless fanout instrumentation to preserve real-time chat ordering and horizontal scale (accept rate 99.39→99.99%, ack p95 767ms→6.6ms)

Serverless Architecture & Cost Optimization

- Batch-to-EDA migration expanding throughput 270x, Lambda hybrid pipeline design
- BigQuery pricing model analysis with read-method switch — 82% cost reduction

Legacy Modernization & Reliability

- Express monolith to NestJS via 3-stage progressive migration, TDD-driven incident-free deployments during major releases
- Shared package system unifying 7 projects, cutting deployment from 3h to 15min

Engineering Productivity & Quality Verification

- Established team quality gates that verify AI-assisted development through TDD, integration tests, review checklists, and Git hooks
- Automated pre-review domain-knowledge and rule checks through Git-based shared context plus local RAG/MCP search
- Presented an Amazon Q Developer-based data pipeline scaling case at Games on AWS 2025

EXPERIENCE

GameDuo — DEV Team Platform Part Leader

2025.01 - Present

Mobile game development and publishing company. Shared backend platform design for 6 live game environments (12 operating branches), data pipeline engineering, and common package system operations

DEV Team Platform Part Leader (2026.04 - Present)

- After the Platform Part was formed, prioritization and verification criteria for shared backend work were fragmented across games. Standardized platform work from backlog to verification to completion with task-type criteria, review gates, and templates. Established an initial operating system for tracking shared infrastructure, package, and chat-server work under the same criteria.
- After AI-assisted development adoption, output quality and verification responsibility depended on each engineer's usage style. Defined a five-dimension sizing framework with TDD/review-based verification principles across scope, complexity, uncertainty, collaboration, and validation. Established the basis for evaluating deliverables by tests and reviews regardless of AI-tool usage.
- Development-automation productivity varied significantly by team member, while individually discovered context and practices did not propagate, causing repeated trial-and-error. Chose Git-based config-as-code over individual documentation — docs must be re-read by humans, but codified config integrates directly into the runtime and applies automatically. Built a shared team development-context platform unifying principles, coding standards, domain knowledge, specialist agents, and hooks in one git repo. Enabled all team members to start with the same AI session context via git pull — converting shared rules, domain knowledge, and specialist agents into searchable team assets.

DEV Team Server Developer (2025.01 - 2026.03)

- Marketing data query cost surge (full-scan billing model). Evaluated partitioning but rejected due to Write IOPS increase; adopted gRPC distributed reads (Storage Read API). Reduced BigQuery read cost per TB by 82% ($6.25→$1.1).
- Batch processing collection limits (60 days). Migrated from batch processing to EDA-based on-demand pipeline — distributing peak load and enabling throughput scaling. Expanded metrics range 6x (60→360 days), processing time 2h→5min.
- Code duplication and convention inconsistency across 7 projects. Replaced manual code sync with shared NestJS package system (10 utility modules + Game Server Kit, auto-deploy). Automated upgrades cutting deployment time 3hrs→15min.
- Marketing query latency (18s) from fragmented ad platform data (Google Ads/Meta/TikTok). Normalized read path + index tuning to avoid full-scan on 100-column table. Performance improved 97% (18s→0.5s).
- Regulatory risk from game probability disclosure requirements. Integrated DynamicModule-based probability package across multi-game projects + CDK-based audit log pipeline. Achieved 94%+ integration test coverage (12 suites, 115 tests).

PROJECTS: Real-Time Game Chat Server (Multi-Tenant), Marketing Integrated Platform, Internal Common Library System

JNPMedi — Dev2 Team Docs Squad Backend Engineer

2023.06 - 2024.08

Clinical trial data management solution startup with KRW 16B cumulative investment (MSA-based)

- 5+ monthly e-signature failures and clinical data-integrity risk caused by deadlock patterns. Reproduced and diagnosed FK shared-lock deadlocks, then adjusted transaction boundaries to prevent recurrence. Eliminated 5+ monthly e-signature processing failures.
- VDR project schedule delay under MVP launch pressure. 3-week emergency sprint (2 BE + 1 FE), Event-Driven PDF conversion + permission-based workflow. MVP launched on schedule.
- Delayed email failure detection (hours). Constructed SES+SNS event pipeline. Bounce/complaint detection cut to seconds, Slack instant alerting (published tech blog).

PROJECTS: Maven Docs System Reliability Improvement, Maven VDR MVP Development

Medgo — Dev Team Backend Engineer

2022.04 - 2023.06

Telemedicine and medication delivery platform startup with 300K cumulative downloads

- Regression risk increased as auth, appointment, and notification changes accumulated around an Express.js single-file app.js structure. 3-stage progressive migration (single-file→layered→NestJS) + JS→TS. TDD 80% coverage, zero incidents during major releases.
- Manual prescription entry bottleneck (3-5min per case). Naver Clova OCR + MFDS (Korea FDA) API integration for full workflow automation. Pharmacist processing time cut 90% (3-5min→30sec).
- Status reflection delay in consultation/dispensing/delivery (tens of seconds). Socket.IO real-time sync + Web Push notifications. Reduced to under 1 second, eliminated patient wait complaints.

PROJECTS: Doctor/Pharmacist Web Service Enhancement, Backend Architecture Migration

SimpleHAN — Dev Team Backend Engineer

2021.07 - 2022.03

Custom software consulting firm

- Manual learning management (attendance, completion, progress tracking). Built LMS full-stack (QR attendance, PDF certificates, Excel bulk registration). Digitized operations for 300+ students per semester.
- New matching and counseling features had to be added to a legacy PHP system, while direct changes carried high regression risk. Integrated an independent Spring Boot app through a constrained iframe/postMessage boundary and implemented a weighted-average matching algorithm. Launched the JOB Agent service while isolating the legacy-system impact surface.

PROJECTS: Learning Management System (LMS) Development

Personal Projects (Side) — Platform / Tooling Engineering

2026.04 - Present

Self-directed platform/tooling engineering outside work — development workflow automation framework and side projects

PROJECTS: Public AI CLI Runtime Launcher

TECHNICAL SKILLS

Backend: Node.js, TypeScript, JavaScript, NestJS, Express.js, TypeORM, Java, Spring Boot, Jest, TDD, REST API, gRPC, Testcontainers, Integration Test, OAuth2 — Legacy-to-NestJS progressive migration, shared package system establishment, TDD-driven development, REST API / gRPC design

Database: MySQL, BigQuery, ElastiCache (Redis), Redis, Redis Streams, Valkey, ClickHouse — Query optimization, deadlock root cause analysis, large-scale table normalization, Redis distributed locking

Cloud & Infra: AWS (Lambda, ECS/Fargate, S3, SQS/SNS, CDK, EventBridge, Kinesis, Athena), GCP (BigQuery, Pub/Sub), Docker, GitHub Actions, CI/CD, Terraform, ARM64 (Graviton) — Batch-to-serverless migration, EventBridge/Kinesis event pipeline design, BigQuery cost optimization, CDK-based infrastructure as code, CI/CD pipeline

Quality & AI Verification: TDD Governance, Integration Test, Code Review, Git Hooks, Evidence Search, AI Output Verification — AI-assisted outputs verified through TDD, integration tests, review gates, Git hooks, and evidence search

EDUCATION

Inha Technical College

Bachelor's (Advanced Major Program, Concurrent Employment), Computer Information Engineering

2022.02 - 2023.02

Inha Technical College

Associate (Concurrent Employment), Computer Information

2017.02 - 2022.02

CERTIFICATIONS

SQLD (Structured Query Language Developer) — Korea Data Agency, 2025.12

Engineer Information Processing — Human Resources Development Service of Korea, 2023.06

PRESENTATIONS & WRITING

Boosting Developer Productivity with Amazon Q Developer

Conference Talk — 2025.10

Games on AWS 2025 Customer Session

Email Delivery Monitoring through AWS SES Event Logs

Internal Seminar → Tech Blog — 2024.03

Presented internally then published as tech blog

TRAINING PROGRAMS

Hanghae Plus Backend 6th

2024.09 ~ 2024.11

Hanghae99

TDD and Clean Architecture, Concurrency Control, Redis

Week 1 Hall of Fame, Black Badge Completion

OPEN SOURCE

oh-my-openagent: Fix cached model context limits for Anthropic providers

2026.03

opencode: Fix macOS sidecar process hang on desktop app

2026.02

CodexBar: Support kiro-cli 1.24+ Q Developer format

2026.02

PROFESSIONAL DEVELOPMENT

System Design Interview Study

2025.08 ~ 2026.04

System Design Interview: An Insider's Guide Vol. 1 & 2

Node.js Design Patterns Bible Study

2025.10 ~ 2025.12

Node core concepts, concurrency, event loop, module system